FSD: A novel forged document dataset and baseline

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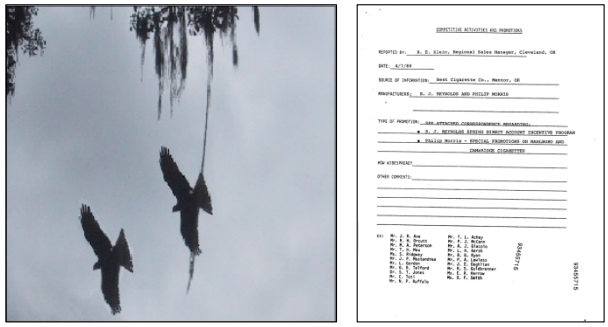
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***Abstract*— Forgery detection in multimedia is an emerging research area in the field of digital content security. The current research focuses on finding several solutions to digital image forgery detection but a small number of works have been reported for scanned document forgery detection. The reason would be the unavailability of a forged scanned document (FSD) dataset. A scanned document can be manipulated in different ways: deleting a region, copy paste a region of the document either in the same document or in another document. Detection of these manipulations becomes arduous when a post-processing operation is performed on the manipulated region. To advance research in this field, we created a large dataset for the FSD. To generate such a dataset, we collected scanned documents from the FUNSD dataset (Publicly Available). The resolution of each sample of the dataset is of size 755x1000. The samples of the dataset are forged using different post-processing operations. The dataset comprises 6656 instances of manipulated scanned documents with their ground truth masks. To build the baseline for forged scanned document detection, we evaluated state-of-the-art object segmentation algorithms on the FSD dataset. Experiments demonstrate that the FSD well represents forged scanned documents and is quite challenging.**

Keywords—Security, Digital Document, Authentication, Deep Learning

I. INTRODUCTON

Forgery detection in scanned documents refers to localizing the forged region in the digital document. Localization of the forged region in scanned documents is almost like localization of forged region in digital image and object segmentation of computer vision. However, the properties of the forged digital image are totally different than the properties of a forged scanned document. Thus, a dataset of forged digital images (natural scenes) can't be used for the evaluation of forged scanned documents. A visual difference between a forged digital image and a forged scanned document is shown in Fig. 1. The digital image has foreground objects and a background. They are of different colors, shapes, and different edges of the objects, etc. However, in scanned documents, there are texts which are of very small edges and with dense information. So, they both are very different in nature with their properties and features. Also, a method to detect forged regions in a digital image or detection of an object in an image can't be directly applied to detect forged regions in a scanned document. The reason is obvious and understood by an example: a little child who doesn't know, how an apple or pear looks like. What is the color and shape of an apple or pear? Then he/she can't differentiate between an apple and a pear.



(a) (b)

Fig. 1. Difference between (a) forged digital image and (b) forged scanned documents

A limited number of solutions have been reported on document forgery detection [1]–[5]. Most of these solutions are statistical or conventional machine learning-based methods and validated on a very small number of samples. The cause is the lack of availability of large-scale datasets. In addition, the current research in any field focuses on deep learning techniques which require datasets of large scale. The verification of learning models trained on a small-scale dataset will lead to overfitting. Thus, we can summaries the following challenges,

- Most of the techniques are based on the assumption that tampered regions have some distortion (i.e., noise or blur). However, if a forger is using a high-quality scanned document and copies a region without any post-processing operation, there is a possibility that tampering processes don't introduce any distortion.
- Techniques are not trained on the accurate dataset. For example, MIDV-500 [6] dataset is for identity documents that are not scanned document datasets.
- Existing works are not trained or validated on forged scanned document datasets. This results in many false positives.

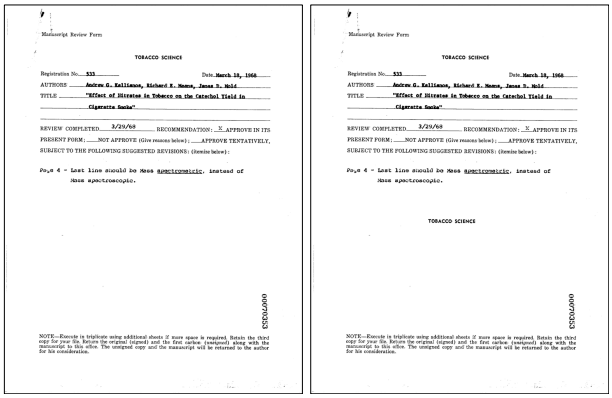
Except for the aforementioned problems, detection methodologies have challenges like post-processing operations on copy-move or spliced forged regions. Internal properties and features of scanned document images are different from features of natural images captured using digital cameras. Therefore, it stands to reason that forged region detectors learned from natural images or identity documents are not suitable for scanned document images.

To overcome the above challenges in forged document detection and for wide and in-depth research we generate a large dataset dedicated to forged documents only. We collect some original scanned documents from FUNSD [7] dataset and use Adobe Photoshop [8] to generate a forged document with their annotated ground truth mask. The original scanned documents are printed documents on an A4 sheet which are scanned through the scanner and collected as FUNSD dataset.

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We observe that forgery is possible in multiple ways on scanned documents too. Based on this observation, we tamper scanned documents in the following ways-

- Copy-Move Forgery- In this type of document forgery, a region of a document is copied and pasted into the same document intentionally to hide any information about the document or to misguide someone with false information.
- Splicing of document content: The content of one document is copied and pasted to another document.
- Resampling of information: Information is resized in the form of either downscaled or upscaled.



(a) (b)

Fig. 2. Example of (a) Authenticated and (b) Forged Documents

Along with these forgeries, four different pre-processing/post-processing operations and three different geometric transformations have been carried out over the forged documents. An example of forged and original documents is shown in Fig. 2. Details of operations implemented over the forged region using Adobe Photoshop are given in the third section of this paper. The dataset contains 6656 images including all forged documents and their ground truth masks. The main contributions of this work are-

- To the best of our knowledge, there is no large dataset available for forged document detection. The generated dataset is having a wide variety of forged documents including multiple types of forgeries, pre-processing and post-processing operations on forged regions. That can be used to train and validate forged document detection models.
- We also benchmark state-of-the-art object segmentation deep learning models on a generated dataset, which can be used as a baseline for future algorithm development.

Additionally, the generated dataset can be used to develop an algorithm to secure scanned documents by utilizing the properties of different categories of forged documents.

The second section of the manuscript discusses motivations and related works. The third section gives a detailed discussion about the annotated dataset and how it is generated. The fourth section defines experimental work and a baseline on the proposed dataset. The work is concluded in the fifth section with future research direction.

II. Motivations

The importance of datasets can't be overstated in the case of data-driven techniques [9]–[14]. It is the task of a deep learning or machine learning model to learn features from a large variety of instances of a dataset. When it comes to the object segmentation task of computer vision area there are many datasets available in the public domain. These large datasets like [15], [16], and [12], [13] are being actively supported in the research community. If we talk about the datasets of forged images, then datasets like CoMoFoD [17], CASIA [18], IEEE-IFS [19], and CUD [20] are very remarkable datasets, which are being used in abundance in the research community.

There is a scope for improvement in the research of detection and localization of forged documents which is affected due to the unavailability of a dataset. According to the forensic experts group [21], alteration to the pages or content of the document includes an example of tampering with the document. Since documents are highly used for verification and authorization, the increased use of tampering in such contents questions the integrity of such content. Hence, the detection of the altered region in the scanned document is a need of the world. Therefore, the unavailability of a scanned document dataset and detection benchmark being close to real-world application motivated us to work in this research direction.

A good benchmark dataset should have the following five characteristics:1) A large number of instances to train and validate a detection model. 2) Proper annotation of the forged region in the ground truth mask. 3) Different categories of forgery. 4) Each category (type of forgery) should have enough instances and 5) Instances should have pre-processing and post-processing operations on the forged region that make it approach real-world application.

A dedicated dataset for identity documents is developed by Arlazarov et al. in 2019 [6] called MIDV-500. It is having 500 video clips for 50 different identity document types with ground truth. Similarly, Sidere Et al. generate a dataset in 2017. This was again for only identity documents. A method for text line forgery is developed by Van Beusekom J et al. [22]. This methodology was evaluated on five different types of datasets which is generated from different printers and at different depths. Although, these datasets are not available in the public domain which lags the research in this domain. In 2021, a new dataset is developed by Mahfoudi et al. by introducing copy-move forgery in payslip documents. TABLE I gives a comparison of an already reported dataset and generated dataset in the field of forged document detection. Here abbreviations are-

- PA: Publically Available (Y: Yes, N: No)
- Inst.: Number of Instances
- Res.: Resolution
- DT: Document Types (ID: Identity Documents, PSD: Printed Scanned Documents, HSD: Handwritten Scanned Document)
- FT: Forgery Type (TA: Text Addition, CP: Copy-Paste, SP: Splicing, R: Resampling)
- LT: Level of Tampering (W: Word Level, R: Region Level)

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TABLE I. COMPARISON OF FSD (PROPOSED) AND ANOTHER RELATED DOCUMENT DATASET

Dataset	PA	Inst.	Resolution	TD	FT	LT
MIDV-500 [6]	Y	500	1080 × 1920 px	ID	-	-
TP300 [22]	N	43	2480 × 3507 px	PSD	TA	W
PPC300 [22]	N	49	2480 × 3507 px	PSD	CP, S	W
TPLJ [22]	N	64	2480 × 3507 px	PSD	CP, S	W
TPCLJ [22]	N	50	2480 × 3507 px	PSD	CP, S	W
DTL [22]	N	196	2480 × 3507 px	PSD	CP, S	W
CIMD [23]	Y	1197	1342 × 943 px	PSD	CP	R
SPD [24]	Y	477	2480 × 3507 px	ID	CP, S	R
HDD [25]	Y	500	2480 × 3507 px	HSD	CP, S	R
ACPR [26]	N	200	-	HSD	CP	R
FCD [27]	N	600	-	PSD	CP, S	W
FSD (Proposed)	**Y**	**6656**	**777 × 1000 px**	**PSD**	**CP, S, R**	**R, W**

Most of the datasets have a smaller number of instances. These datasets have limitations- number of instances, type of forgery, or unavailability in the public domain. The published datasets are in the form of handwritten or identity documents and these datasets scanned can't be used for the printed scanned document as they give a piece of wrong information related to the forged region or forged document. Keeping all these limitations and challenges, we decided to develop a dataset that should have a large number of instances with different types of forgery.

III. ANNOTATION OF FSD DATASET

A. Image Collection

Deep learning models mimic human brains and almost depend on training. So, the overall dependency of deep learning models is either on the dataset or the architecture. A dataset of forged document data is very important to train the model and to the best of our knowledge, none of the forged digital document datasets exists online in the public domain (from where a model can learn). Hence, a dataset having a different type of forgeries is constructed. The **FSD dataset**[1] can be accessed after taking permission from the authors. Some pre-processing operations and geometrical transformations have been applied to the forged region in documents which effectuate the multitudinous of this dataset. This dataset is generated from the existing FUNSD dataset, which is dedicated to text detection, optical character recognition (OCR), spatial layer analysis, and form understanding. This dataset contains 149 documents for training and 50 document images for testing purposes. From these documents, we have fabricated 3284 forged documents and picked out 144 original documents as it is for the non-tampered document. All presented documents and ground truth in the constructed dataset are in png format. Dedicated directories are created for the training and testing of the model. One more directory is created for authenticated documents that are not tampered with in any way and can be used to validate the model with non-tampered documents also. The directory tree structure is shown in Fig. 3 for the dataset which explains the number of documents and ground truth and location of documents in the dataset with their directory names.

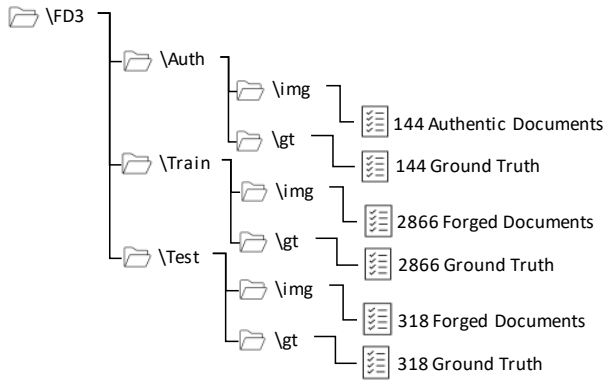


Fig. 3. Tree structure of directory and content of the constructed dataset

B. Category of Forgery

Three types of basic forgeries have been constituted to construct a dataset: (i) Copy-Move Forgery- In this type of document forgery, a region of document is copied and pasted in the same document intentionally to hide any information of document or to misguide someone with false information. (ii) Splicing of document content: The content of one document is copied and pasted to another document. (iii) Resampling of information: Information is resized in the form of either downscaled or upscaled. Along with these forgeries, four different pre-processing/post-processing operations and three different geometric transformations have been carried out over the forged documents. Details of operations implemented over the forged region using Adobe Photoshop are as follows (measure for post-processing/pre-processing operations are used of Adobe Photoshop Application):

- First is a simple forgery in which a region is copied from one document and pasted to the same document itself (copy-move) or another document (splicing) without any pre-processing or post-processing operations.
- The second is brightness enhancement of the forged region of the document up to 100 points on the scale.
- The third post-processing operation is contrasted stretching up to 100 points on the scale. This operation is implemented in a forged region like brightness enhancement.
- The blurring of the forged region is the fourth operation that is brought out up to two neighboring pixels.
- In the fifth operation, white Gaussian noise is added up to 20% to the forged region of the document.
- Except for these image processing operations, three geometrical transformations are carried over the forged region. Among these three operations, the first transformation is translation.

[1] https://drive.google.com/file/d/1mCFH41b-v4OXeYAVFB2MppSQKa-BHtLH/view?usp=sharing

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- The second geometric transformation is the scaling of the forged region. This operation accomplishes the definition of the third type of forgery which is the resampling of information.
- The last operation is the rotation of the forged region which is rotated up to 10° in the forged document.

TABLE II. DETAILS OF CONSTRUCTED DATASET

S. No.	FT	Resolution (h x w)	Operations	No.	No.	Total
1	Copy -Paste	1000 × 755 to 1000 × 777	*Simple*	199	199	398
2			*Brightness Enhancement*	199	199	398
3			*Contrast Stretching*	199	199	398
4			*Noise Addition*	199	199	398
5			*Blurring*	199	199	398
6			*Translation*	199	199	398
7			*Rotation*	199	199	398
8			*Scaling*	199	199	398
9	Splicing	1000 × 755 to 1000 × 777	*Simple*	199	199	398
10			*Brightness Enhancement*	199	199	398
11			*Contrast Stretching*	199	199	398
12			*Noise Addition*	199	199	398
13			*Blurring*	199	199	398
14			*Translation*	199	199	398
15			*Rotation*	199	199	398
16			*Scaling*	199	199	398
17	Non-tampered	1000 × 755 to 1000 × 777	*Simple*	144	144	288
			Grand Total:	**3328**	**3328**	**6656**

TABLE II gives the overall detail of the dataset with the number of images in the dataset. Fig. 4 shows a visual demonstration of a forged document having all operations over the forged region and their ground truth masks.

C. Dataset Splits

It is important to distribute the dataset into training, validation, and test set. To ensure the training, validation, and test result, we distribute the dataset into two parts- one is the training set, and another is the test set. The training set can be further divided into training and validation sets based on the algorithm's criteria. So, it is a researcher who is responsible for the further division of training sets into training and validation sets. We are providing the dataset in the public domain. We evaluate the dataset on a high computing server and set a benchmark on predefined object segmentation models.

IV. EVALUATIONS AND BASELINE

For the evaluation of the proposed dataset and to set a baseline, we take different object segmentation and deep learning models. On these standard deep learning segmentation models, we train our proposed dataset and find the training and validation evaluation results. For the evaluation and benchmark, a high-performance computing server is used. A continuous job pool is given to this server where the training of the model is done one by one. We take four different standard object segmentation models. These models are Unet [28], Unet++ [29] and Unet-attention [30]. Python libraries that are used here are- Keras, os, numpy and OpenCV.

For the evaluation of the object segmentation models, the performance measure accuracy and validation accuracy scores are used here. Accuracy with loss value for the training, validation as well as testing of the standard models are used.

Accuracy score can be calculated by the following formula when positive and negative pixels are known:

$$a = \frac{tp+tn}{tp+tn+fp+fn} \quad (1)$$

Except this during testing of the model on separate data IoU score is also calculated:

$$IoU = \frac{tp}{tp+fp+fn} \quad (2)$$

Training results and training loss on different standard models are shown in the TABLE III.

TABLE III. ACCURACY SCORE AND LOSS VALUES DURING TRAINING OF DIFFERENT STANDARD MODELS

Model	Accuracy		Loss	
	Train	***Validation***	***Train***	***Validation***
Unet	0.9847	0.9784	0.0493	0.0931
Unet++	0.9928	0.9919	0.0171	0.0204
Unet-Att	0.9933	0.9927	0.0143	0.0189

Now for the baseline and evaluation of the trained model, we use a test dataset on trained models. The test dataset is separated from the training data and is neither used as train data nor as validation data. Now coming to the baseline, the performance of the trained models on test data is shown in TABLE IV. We use Unet, Unet++ and Unet-attention trained models to validate them on the test set as they give better training IoU scores rather than others.

TABLE IV. IOU SCORE OF TEST SET ON DIFFERENT TYPE OF FORGERY OF DIFFERENT TRAINED MODELS

Model	Type of Forgery	Accuracy	IoU Score
Unet	Copy-Move	0.9606	0.4702
	Splicing	0.9617	0.4522
Unet++	Copy-Move	0.9708	0.5455
	Splicing	0.9693	0.5275
Unet-attention	Copy-Move	0.9709	0.5522
	Splicing	0.9714	0.5509

The higher the IoU score, the better the generalisation capability of any segmentation model. Although the basic models Unet [28], Unet++ [29], and Unet-attention [30] achieved improved accuracy, they lacked the ability to segment the faked areas more effectively. By looking at the IoU ratings of these models, this is clear. Due to the complex characteristics of scanned documents, you may thus get better training accuracy, but you may not achieve superior segmentation results on the validation test sets.

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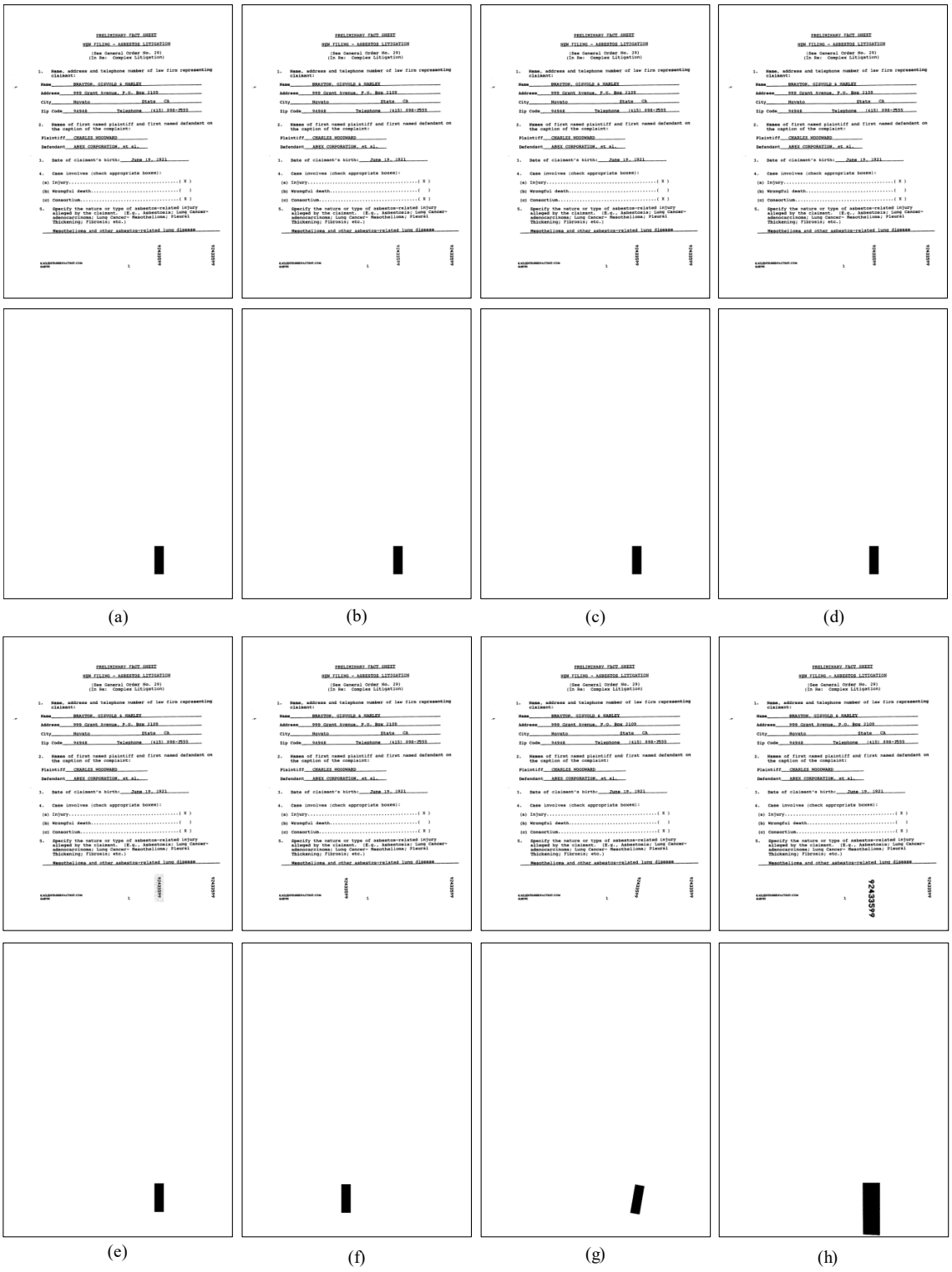


Fig. 4. Operation-wise forged documents and their ground truth mask (a) Simple (b) Brightness Enhancement (c)Contrast Stretching (d) Noise Addition (e) Blurring (f) Translation (g) Rotation (h) Scaling

V. Conclusion and Future Work

The current research on digital content security focused on detecting forged images, whereas very limited works have been reported in the literature for forged document detection. Such works evaluated their model's performance on a very small scale of forged document datasets. But the model can overfit if they are trained on a small dataset. Thus, to address such issues, this paper proposed to develop a largescale dataset for a forged document which will be available with the link given in the footnote on request. Several challenges are also included in the dataset. These are in the form of post-processing operations on the forged region such as noise addition, rotation, scaling, brightness change and contrast adjustment. The proposed dataset contains a total sample of 6656, out of which 60%, 20% and 20% of the dataset are considered for training, validation and testing respectively. Some of the state-of-the-art approaches have been coded and evaluated on the proposed dataset to find out the criticality of the proposed model. The highest accuracy on the proposed

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dataset is obtained by the Unet-Attention whose IoU is 0.55. However, there is ample scope to develop more sophisticated models on the proposed dataset. Future work not only will focus on increasing the sample sizes further but also focus on proposing a new deep architecture to increase the performance of forged document detection. State-of-the-art-techniques are there for the natural images [31-34]. These techniques can also be used and compared with the baseline methods.

DATA AVAILABILITY STATEMENT

The dataset that supports the findings of this study is available on request from the given link on the footnote of the second page of the manuscript.

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